LUKE THOMPSON

FIRMWARE & EMBEDDED SOFTWARE DEVELOPER

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Portfolio: xylardark.github.io/Portfolio-Studio/u/admin/miovision · GitHub: github.com/XylarDark

PROFESSIONAL SUMMARY

Developer with hands-on embedded experience: a 12-month contract at TruckSail implementing STM32/CAN firmware and finding defects for reliability, alongside the JS/SQL web side of the product. Computer Systems Technology diploma (Software Development), Mohawk College, 2019. Since February 2026 I've written C++ and Python on HomeWorld, my solo Unreal Engine 5.8 project, including test and build automation that won't report a pass it can't prove. Hamilton, ON; open to hybrid in Kitchener.

CORE SKILLS

Embedded	STM32 microcontroller firmware, CAN bus, defect finding for reliability
Languages	C++, Python, C#, JavaScript/TypeScript, SQL, PowerShell
Test & build	Automated test runners, pytest, Unreal Automation tests, build scripting, Git/GitHub, GitHub Actions
Also	REST APIs, SQL Server, MySQL, NGINX basics, AI-assisted development with review gates

EMBEDDED EXPERIENCE

Software Developer · 12-Month Contract · TruckSail · Ancaster, ON 2020

- Implemented STM32/CAN firmware with defect finding for reliability.
- Built JS/Bootstrap UIs and optimized SQL for the web side of the product.
- Maintained GitHub workflows; cross-browser QA and password-management security support.

RECENT C++ AND PYTHON WORK

HomeWorld — Unreal Engine 5.8 / C++ game and tooling Feb 2026 – present

Independent (solo) · Unreal Engine 5.8, C++, Python · private repository, available on request

- C++ gameplay and systems: character movement, Gameplay Ability System, a versioned save system with per-system slices and migrations that refuse unknown newer saves.
- A C++ editor module with commandlets (UCommandlet, package save, PCG and MassEntity APIs) for unattended setup tasks.
- A headless test runner that hard-fails on five causes of fake passes, including stale binaries and a wrong engine version, and a build wrapper that retries lock failures and queues builds machine-wide.
- 42 C++ automation test files and 150+ Python test files.

Shift — playable shop & delivery game Aug – Sep 2026

Independent · Phaser 3, TypeScript, Vite · Live: xylardark.github.io/ShiftGame

- Fixed 60 Hz timestep simulation with render interpolation and a high/mid/low render budget for phones; 83 Vitest test files.

OTHER EXPERIENCE

Video Game Developer · Contract · OutFusion · Hamilton, ON 2022

- Built and tested C# REST API integrations for player data; integrated and tested game UI across mobile and desktop; bug verification before release.

Delivery Driver (Uber; Kindling Cannabis) · Hamilton, ON 2022 – 2026

- Independent schedule while building a software portfolio in 2026 (HomeWorld, Shift, and Portfolio Studio). Reliable, accurate delivery in regulated retail settings.

EDUCATION

Diploma, Computer Systems Technology — Software Development · Mohawk College · Hamilton, ON · 2019

Valid Ontario Class G licence · Open to hybrid at the Kitchener headquarters (commuting from Hamilton)